



Atul Kumar

Software Developer | 8800575522 | atulkumarit05@gmail.com

Summary

Skilled software developer with expertise in PHP, Frameworks, NodeJS, Python, MySQL, WebRTC and DevOps, delivering scalable and efficient solutions.

Designed and implemented dashboards, video conferencing applications, and video streaming systems using PHP, NodeJS, WebRTC, DevOps and FFmpeg.

Optimized API performance through multithreading and reduced execution time for enhanced responsiveness.

Migrated and optimized projects to remove Wowza dependency, reducing costs and improving system performance.

Developed RESTful APIs and integrated real-time communication tools like WebRTC and WebSocket. Configured and deployed solutions in cloud environments, including AWS, for high-quality streaming and scalability.

Provided seamless client support and delivered monthly code updates with a focus on quality and efficiency.

SKILLS

Programming Languages:

PHP (Laravel, CI, OpenCart, Symphony), WebRTC, NodeJS, HTML, CSS, JavaScript, jQuery knowledge of GStreamer, C/C++

Other: DevOps (Docker, Jenkins, AWS, Ansible, Terraform), MySQL, RTMP, MongoDB, FFmpeg

Main competence areas

- Development method
- IT
- ▢ DevOps

Work roles

- Developer
- Lead Developer

Industry knowledge

- Media
 - Education
 - IT
- Consultancy, Data processing,
Software vendor

Career and projects

Incise Infotech Pvt Ltd (Client NXP)

Apr 2022 - Present - Software Developer (PHP, MySQL, Python, Perl, Shell, AWS, Jenkins, HTML, CSS, JavaScript, jQuery)

Designed and developed new features for an existing dashboard, enhancing both its functionality and user experience. The project involved using PHP and OpenCart for backend development, along with JavaScript, HTML, and CSS to create a dynamic, responsive, and user-friendly frontend. MySQL was utilized to manage and query the database, ensuring efficient data storage and retrieval.

Uploading and sanitizing data using Perl involves ensuring that data is safely received and processed before being stored or used. The process typically includes validating input data, checking for malicious content, and removing or escaping special characters to prevent security vulnerabilities.

Creating a test script by using python for testing involves developing automated tests to ensure the functionality, performance, and security of the application.

Deploying code using Jenkins involves automating the process of building, testing, and deploying applications to various environments (e.g., development, staging, production). Jenkins, an open-source automation server, is commonly used to streamline continuous integration (CI) and continuous deployment (CD) workflows.

Gradestack Learning Pvt Ltd

Oct 2021 - Jan 2022 - Software Engineer L1 (WebRTC, Go, Redis, ffmpeg)

Developed WebSocket-based video streaming solutions for high-performance delivery.

Created a chat application using Node.js and MongoDB, facilitating real-time communication.

Built APIs for Android app integration, ensuring seamless functionality.

Published videos from multiple sources, ensuring adaptive playback using OBS studio create a live event and publish on server using RTMP.

Optimized internal projects by integrating Redis for efficient data caching.

Collaborated with designers and QA teams to improve the UI/UX of internal tools.

Saregama Pvt Ltd

May 2021 - Oct 2021 - Deputy Manager (PHP, WebRTC, MySQL, ffmpeg, HTML, CSS, JavaScript, jQuery)

Enhancing the application by using CodeIgniter (CI), HTML, JavaScript, and MySQL to improve performance, scalability, and user experience

Developing a robust video conferencing application using RTMP to run campaigns efficiently, using Node.js, MongoDB, PHP, HTML, CSS, and JavaScript, and deploying on AWS for scalability and performance.

The video frames were extracted and converted into images for the presentation video. This process involved capturing individual frames at specific intervals to ensure clarity and relevance. By using tools like FFmpeg, high-quality images were generated, maintaining the original video resolution. These extracted frames can now be used for presentations, analysis, or further editing, enhancing the visual representation of key moments from the video

Ashmar IP

Dec 2018 - Dec 2020 - Software Developer (WebRTC, PHP, Laravel, MySQL, ffmpeg, AWS, DevOps, HTML, CSS, JavaScript, jQuery, MongoDB)

Developing a video streaming and recording application by replacing the Wowza service with Janus WebRTC. The application will enable real-time video streaming and user video recording using Janus, leveraging WebRTC for low-latency communication. It will be built using AWS Server, Node.js, MongoDB, HTML, CSS, JavaScript, and jQuery, ensuring seamless integration with existing web applications. The system will support live broadcasting, on-demand playback, and secure storage of recorded videos.

After completing a campaign, generating the report takes approximately 5–6 hours because we analyze facial expressions. To reduce processing time, I optimized the workflow using Python multithreading. By parallelizing facial expression analysis, the system can process multiple video frames simultaneously, significantly improving performance. Additionally, I implemented efficient data handling and optimized algorithm execution to further enhance speed and scalability. As a result, the report generation time was reduced, making the system more efficient and responsive

Upgrading the application from Smarty to Laravel involves a complete transition from the traditional Smarty templating engine to Laravel's modern MVC framework. This migration enhances maintainability, scalability, and security while improving performance.

SCICOM

May 2017 - Dec 2018 - Software Developer (PHP, MySQL, HTML, CSS, JavaScript, jQuery)

Creating a school management software from scratch using PHP, MySQL, HTML, CSS, JavaScript, and jQuery involves planning, designing, coding, and providing ongoing client support. The process starts with gathering requirements, followed by designing the system architecture and database. Then, the front-end is developed using HTML, CSS, JavaScript, and jQuery, while the back-end is powered by PHP and MySQL. After coding, the software is deployed, and client support is provided through training, and ongoing assistance

Developing a comprehensive placement portal from scratch for students and placement agencies using PHP, MySQL, HTML, CSS, JavaScript, and jQuery. The portal will facilitate student registrations, profile creation, resume uploads, and job applications while allowing placement agencies to post job openings, manage applications, and shortlist candidates. It will include an admin panel for user management, job listings, notifications, and analytics to track placement activities. The system will ensure a seamless and interactive experience through responsive design, AJAX-based interactions, and secure authentication mechanisms

Microtek International Pvt Ltd

Jul 2013 - Jun 2014 - Coordinator

Assigned and closed user complaints involve efficiently managing and resolving issues raised by users to ensure their satisfaction and maintain product quality.

Prepared reports on task completion and presented them to senior management.

Maintained clear communication channels between teams to enhance collaboration.

Education, courses and certifications

Education

2017	Ajay Kumar Garg Engg College Ghaziabad B-Tech, Computer Science
2012	N.I.E.T Modipuram Meerut Polytechnic, Information Technology
2009	Janta Inter College Kaithwari Meerut 10th

Language

	English
	Professional working proficiency